



A multimodal meta-learning-augmented EmoTriSense for emotion recognition

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Abstract

Emotion recognition is an important issue in healthcare, education, and human–computer interaction because correct recognition of different emotional conditions could dramatically improve the efficiency of applications. Nevertheless, the current methods have been limited by the lack of full utilization of modality-specific mechanisms, poor inter-modality cohesion, and lack of adaptability in different tasks. The traditional fusion models tend to be based on naive concatenation, that is, they do not reflect contextual dependencies between audio, video, and electroencephalography (EEG) signals, as a result of which they generally do not generalize optimally and are not especially efficient. Multimodal Meta-Learning-Augmented EmoTriSense System is proposed as a holistic system of multimodal emotion recognition. It takes audio, video and EEG modalities and processes them separately and then combines them using a Cross-Modality Relational Attention (CMRA) fusion process. Whereas direct concatenation does not train the relational mappings, CMRA can be trained to learn such mappings allowing audio cues to strengthen the visual signal and the EEG data to contextualize the auditory and visual signals. The modality pipelines include audio decoding done using transformers, optical flow stabilization using Retina Net face detection and a 3D Convolutional Neural Network (3D-CNN) with attention, as well as Multiscale Temporal Dynamic Convolution (MSTDC) with multiscale feature selection. Such sophisticated representations are further improved through meta-learning that includes an episodic memory module, which makes them adapt strongly to tasks. Experimental findings highlight the effectiveness of EmoTriSense, with 95.17, 96.76, 95.48, and 93.95 accuracy on the DEAP dataset (valence, arousal, dominance, liking), using only 66 s of running time, 48.6 memory usage of the CPU, and 0.5 M parameters. On the EAV dataset, the performance rises to 98.8, 99.2, 99.0 and 97.5, thus demonstrating the effectiveness of its modality-specific pipelines, CMRA fusion, and meta-learning platform of scalable multimodal emotion recognition.

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1 Introduction

The systems of emotion identification are becoming an inevitable part of the dynamically developing spheres of artificial intelligence and human–computer interaction (Du et al. 2022). Emotion recognition application will be used to support the development of human–computer interaction (Rajpoot and Panicker 2022), individualization of educational processes, improving customer service, and assisting healthcare professionals (Pandey and Seeja 2022; Iyer et al. 2023). A deeper understanding of the human behavior can be reached by identifying emotions, therefore, allowing more personalized and adaptive interaction in various fields including customer service and mental health assistance (Yu and Wang 2022; Ahmed et al. 2023; Sajjad et al. 2023).

There are various multimodal strategies that were designed in the attempt to explain human affect and they take advantage of audio, visual, and electroencephalographic (EEG) recordings, to determine the affective states. Audio-based affective computing systems (Pan et al. 2023) use advanced speech analytic approaches but this is hampered by differences in prosodic patterns and inherent bias which can result to misclassification. Visual modalities (Lope and Graña 2023) are based on complex facial expression analysis and computer vision techniques, but lack reliability due to inter-individual variation of expression and other contextual ambiguity which does not allow correct inference of emotions. EEG-based recognition of affective states (Singh et al. 2023) has its difficulties due to inter-subject variability in neural dynamics and sensitivity to external noise. Numerous emotion-detection architectures, such as convolutional neural networks (CNNs) (Ma et al. 2019) and deep belief network-glia chains (DBN-GCs) (Peña et al. 2023), multilayer perceptrons (MLPs) (Tzirakis et al. 2017), long short-term memory (LSTM) networks (Alhagry et al. 2017), and capsule networks (CapsNets) (Zhang et al. 2022) have similar weaknesses to multimodal fusion (Chao et al. 2018). These shortcomings include emotional bias, requirement of large labeled datasets in order to train robustly and excessive computational overhead, which prevent a smooth fit to the real world. Besides, the MLP and LSTM models are especially vulnerable to data noise, highlighting the complexity of factors to consider to obtain accurate and fair multimodal affect sensing within various algorithmic models.

Emotion recognition technologies have their drawbacks, even though the potential advantages are high. The issues of potential biases in the algorithms used have raised red flags regarding their justifiable usage. In addition, the risk of misunderstanding emotional conditions will be a substantive issue, which can lead to unforeseen consequences of sensitive fields of recruitment, surveillance, and psychiatric diagnostic practice (Chao et al. 2019a). Practical implementations require situational cognisance and responsiveness to change, which means that these methodologies have to be enhanced to dynamic working situations.

The given approach, which is based on the combination of audio, video and EEG representation, which are obtained out of the DEAP and EAV datasets, is aimed at overcoming the challenges of subjectivity, error in classification, as well as differences

in context and the lack of emotion labels. It systematically examines the capacity of each modality to engage in discrimination and explores ways of combining the above in a multimodal emotion recognition system. The research will be conducted by comparing the outputs of these various methods in a systematic manner and in the process offer a more detailed overview of the complicated area of emotion recognition, thus contributing to a better understanding and developing the discipline.

The proposed methodology has the following contributions:

The proposed Multimodal Meta-Learning-Augmented EmoTriSense System offers the following key contributions:

- Independent processing pipelines for audio, video, and electroencephalography (EEG) data are developed, followed by integration through a Cross-Modality Relational Attention (CMRA) fusion mechanism. Unlike direct concatenation, CMRA learns inter-modality dependencies, enabling audio cues to reinforce visual expressions while EEG signals provide contextual grounding.
- Raw audio waveforms are transformed into expressive representations using vector quantization and transformer-based decoding. This pipeline enhances low-level audio features and contextualizes them for improved discrimination of emotion-related cues.
- A combination of optical flow stabilization and RetinaNet-based face detection ensures stable and accurate preprocessing. 3D-CNN with attention is used for feature extraction, effectively capturing both temporal and spatial patterns in face emotions.
- A Multi-Scale Temporal Dynamic Convolution (MSTDC) framework with multiscale feature selection is employed to capture spatial-temporal dependencies in EEG signals. This design enables the extraction of highly discriminative neural features while reducing redundancy.
- An episodic memory module is incorporated to strengthen adaptability across diverse emotion recognition tasks, allowing the framework to generalize effectively while remaining lightweight and computationally efficient.

Section 2 analyses the relevant work, Sect. 3 explains the suggested approach, Sect. 4 describes the dataset, Sect. 5 presents the analysis and results, and Sect. 6 summarizes the conclusion.

2 Literature review

The related work to emotion recognition using EEG, audio and video data is discussed as follows:

Liu et al. (Liu et al. 2022) have suggested the CEFLNet that is promising in terms of reducing redundancy and complexity in the video sequences. CEFLNet is more right in determining the most informative emotional clips thus enhancing performance of emotion recognition. The implication of this finding is important in the real-life application in emotion-driven technologies and human-computer interaction. However, the method relies on the weak supervision and can be characterized by scalability issues.

Gan et al. (Gan et al. 2022) proposed a DSN-DF model, applied to recognize face emotions using video. Their approach attempts to deal with the complexity of the

expression of emotions in real-life situations by dividing video sequences and deriving spatiotemporal appearance and geometrical features. DSN-DF exhibits a sensitivity to the subtle changes of the facial expression with a high performance noted when it is used in case of controlled datasets. Nevertheless, the implementation of such a strategy on the ground requires a cautious observation of computational requirements, data reliance and even variability that may exist in various datasets.

Jana et al. (Jana et al. 2022) examined emotion recognition using electroencephalographic (EEG) signals by introducing a spatio-temporal frame representation that has the potential in a multiplicity of applications. The proposed methodology incorporates Capsule Networks and Bayesian Optimization, which makes it competitive, particularly in terms of cross-subject inference. However, one should still consider the dependency of the dataset and computational requirements of the EEG-based emotion recognition, which might require additional optimization to be implemented into the real-life context.

Shou et al. (Shou et al. 2022) have carried out empirical studies on multimodal emotion recognition in the conversational situation with quite significant implications to human–computer interaction and sentiment analysis. The contribution of the Dynamic Interactive Multiview Memory Network (DIMMN) is the focus on dynamic interactions and contextual cues in recognizing emotion. The efficacy of DIMMN was found in its superiority over the current methods on benchmark datasets (IEMOCAP, MELD, etc.). However, there are still certain difficulties, such as the lack of data, computational cost and the extrapolation of the model to other contextual environments.

Wang et al. (Wang et al. 2022) explore the combination of various modalities to identify emotions, and it is an interesting study on the Automatic Emotion Recognition (AER). The creation of Multimodal Emotion Database (MED4) is a vital asset to AER research community. MED4 enables the assessment of EEG and speech-based emotion detection, and it focuses on the strength of EEG signals in noisy conditions. Although decision-level fusion strategies are more accurate, such practical considerations as the availability of data and the complexity of computation should be considered.

Wen et al. (Wen et al. 2023) described the new approach of the EEG-based emotion recognition with the use of the STANN. The STANN architecture is very effective in terms of capturing spatial and temporal information that is highly accurate and outsmarts the existing mechanisms on the DEAP dataset. Its cross-modal transfer learning feature and its capability of alleviating the low signal-to-noise ratio of EEG signals highlight the importance of the feature to the field. However, the label dependence of the model and its sensitivity to domain differences introduce real-life challenges that have to be overcome in the practice. Sartipi et al. (Sartipi et al. 2023) also reported the STANN on the identical EEG dataset. Their model addresses the derivation of both the spatial and temporal characteristics by a new hybrid profile. Efficacy of this method is demonstrated by the fact that it performs well on the DEAP dataset, and this indicates its possibility to be applied in practice. Furthermore, the model's adaptability is highlighted by the representations' adaptability to various EEG emotion identification tasks. However, data scarcity and domain sensitivity should be considered, encouraging future work to explore other physiological data modalities.

Yiyuan Chen et al. (Chen et al. 2025) propose LRCOMF framework designed for EEG-based emotion recognition under unlabeled online streaming conditions, where

conventional batch or supervised online methods fail due to non-stationarity and lack of real-time labels. The methodology begins with an offline meta-pretraining stage that learns an initialization capable of rapid adaptation from very limited labeled EEG data, improving generalization to unseen samples. During online deployment, each incoming EEG sample is first evaluated by a sample judgment module, which combines softmax confidence with a prototype-distance-based weighting mechanism to filter unreliable pseudo-labels. High-confidence samples are stored in a multi-task cache and optimized using online meta-learning, enabling the model to learn new patterns while reviewing previously stable samples in a controlled manner. Samples rejected initially are preserved and periodically reassessed through a challenge module, which employs an unsupervised clustering-quality metric to determine whether incorporating them improves representation separability, thereby allowing gradual utilization of harder samples without performance collapse. Although the framework is successful in reducing the effects of errors and successive improvements in performance when there are no explicit labels, a weakness is often observed related to how sensitive its training dynamics and trajectories are to distributional changes over time and label noise. This sensitivity produces unbalanced convergence and increased computational requirements, which can be explained by recurring meta-updates and challenge-induced re-evaluation loops that are caused by real-time operations settings.

Chang Gao and colleagues (Gao et al. 2025) present the UMDA-DDSTGN framework, which is an unsupervised meta domain adaptation approach to EEG based emotion recognition, which addresses dynamic cerebral connectivity modeling and cross subject generalization. This method builds a dynamic directed spatial-temporal graph by using the partial directed coherence to estimate temporally fluctuation and directional information flow between EEG-channels. The following graph is then fed through a Dynamic Directed Spatial-temporal Graph Network (DDSTGN) with graph-attention based on learning spatial dependencies and multiscale residual dilated convolutions based on learning temporal features. In order to achieve cross-subject variability without labeled target data, the framework uses unsupervised meta-domain adaptation, where a meta-adaptive training phase is used to obtain source subject domain-invariant representations via a chain of meta-tasks, followed by a step of unsupervised fine-tuning which is used to align feature distributions between source and unlabeled target domains using covariance-based adaptation. Therefore, the design allows quick adaptation to new subjects with only small quantities of unlabeled data, and strong emotion-discriminative properties. However, one of the major drawbacks of the suggested method is the using of dynamic directed graphs with partial directed coherence that imposes a lot of computational overhead and demands offline pre-processing complexity; together with the meta-learning optimization process, which may slow down the application and growth of the method to practical EEG emotion-recognition systems.

Based on the literature review, some common themes and challenges can be identified in the sphere of emotion recognition using EEG, audio, and video modalities, including the reliance on labelled data; the potential limitation of scalability; the complexity of computations; the sensitivity to the fields of data; the reliance of basic datasets; the limitations of data usage suggested by the real-world situation; and the need to optimise further to suit the field. The literature survey also points at the reliance

on labelled data, possible scaling problems, computational complexity and sensitivity to data domains. In addition to that, data dependency on the dataset, limitations of data availability, and the need to perform further optimisation in the practical world become the typical drawbacks of the presented frameworks. The general overview of the survey is given in Table 1.

Overall, the suggested approach addresses important issues in emotion recognition when combining EEG, audio, and video data by expanding on the discoveries and advancements in the literature study. By fusing innovative techniques with practical considerations, the goal is to advance the area and contribute to the development of more dependable and effective emotion identification systems.

3 Proposed multimodal meta-learning-augmented EmoTriSense system

The current research utilizes audio, video and electroencephalographic (EEG) data of the DEAP data and evaluates the recognition abilities of each of the modalities individually and examines how they can be integrated into a well-rounded system of multimodal emotion recognition. The audio branch starts with the processing of raw waveforms in which low-level features are obtained and later enhanced with contextual modeling. The representations in the form of vector-quantised are in turn decoded using a transformer network to create phoneme units, potentially containing affective data. A meta-learning model is achieved by maintenance of task-specific examples with an episodic memory module and task-specific fine-tuning with emotion-label prediction to optimise the model to specific affective categorisation tasks. The first one is the visual pipeline, which stabilises the frame with the help of optical-flow estimation, subsequently locating facial regions with the help of a RetinaNet detector, and finally obtaining a spatiotemporal feature using a new 3-D convolutional neural network with an attention mechanism. EEG signal processing steps are preprocessing and feature extraction in the form of multiscale spatiotemporal deep convolution (MSTDC) module to ensure the spatial and temporal dynamics are relevant. Multiscale feature selection then forms the most informative attributes resulting in a small size of EEG features that are then part of meta-learning training, and finally, it results in a powerful emotion recogniser. To perform multimodal fusion, the EEG feature vectors, the audio and visual data are first normalised before combining. It uses a Cross-Modality Relational Attention (CMRA) module that is used to model inter-modal dependencies and reinforce complementary representations. The CMRA learns the relationship mappings in such a way, that, auditory signal can be used to supplement visual expressions, and EEG patterns give the contextual basis of both modalities. These fined multimodal representations are then fed into a meta-learning system, and the ultimate affective classification is achieved. The CMRA-like fusion mechanism allows more effective amalgamation of heterogeneous modalities, which promotes a powerful and context-dependent definition of human emotions, as shown in Fig. 1.

Table 1 Overview of the reviewed works

Author & year	Dataset	Methodology	Analysis	Emotion labels	Limitations
Liu et al. (Liu et al. 2022), 2022	BU-3DFE, MMI, AFEW and DFEW	Clip Segmentation, CEFLNet Architecture, Spatio-Temporal Features, Emotional Activation Maps	CEFLNet, a clip aware emotion-rich feature learning network, which effectively reduces redundancy and complexity in video-based facial expression recognition	Anger, disgust, fear, happiness, surprise, Neutral, sadness	Weak supervision, scalability, lack of multimodality, data variability
Gan et al. (Gan et al. 2022), 2022	CK + and MMI	DSN-DF (highlighting the region and intensity, while capturing the speed of change through geometric features)	Effective approach to video-oriented facial expression recognition	Anger, disgust, fear, happiness, surprise	Resource intensiveness, data dependency, parameter tuning, interpretability
Jana et al. (Jana et al. 2022), 2022	DEAP	Spatio-Temporal Frame Group (STF Group), Capsule Network (CapsNet), Bayesian Optimization	Promising spatiotemporal EEG-based emotion recognition, Utilizing the CapsNet architecture-binary classification task of emotion recognition for new data	Arousal, dominance, valence, liking	Limited benchmarking, hyperparameter sensitivity, dataset dependency

Table 1 (continued)

Author & year	Dataset	Methodology	Analysis	Emotion labels	Limitations
Shou et al. (Shou et al. 2022), 2022	IEMOCAP and MELD	Conversational Affective Analysis Model (DSAGCN), Bi-LSTM model to capture historical context information, GCN	Improved Semantic Understanding, recognizes emotions in dialogue contexts	Anger, disgust, fear, happiness, surprise	Ambiguity in emotion labels, dependence on dataset size, class imbalance, hyperparameter tuning, limited model exploration
Wang et al. (Wang et al. 2022), 2022	SEED-IV and DEAP	MED4 database, four baseline algorithms (1-vector + PLDA, TCN, ELM, MLP), Decision-level fusion methods (AVER, WAVER, RF, ET)	Decision-level fusion methods, such as WAVER, prove useful in enhancing emotion recognition accuracy and robustness	Happy, angry, sad, neutral, other	Speech sensitivity to noise, recognition limitations, nonlinear decision-level fusion
Wen et al. (Wen et al. 2023), 2023	IEMOCAP and MELD	Multimodal Fusion for Emotion Recognition, DIMMN, TCN, GRUs and Memory Networks	Comprehensive approach to emotion recognition in multimodal conversations	Happy, sad, neutral, angry, frustrated, excited, disgust, fear, joy	Do not generalize to diverse conversational contexts and emotional expressions, Data Availability, Scalability
Sartipi et al. (Sartipi et al. 2023), 2023	DEAP	STANN features a hybrid design with STE, RAN blocks, with attention mechanism	Robustness of EEG signals, Fusion strategies for improving accuracy, Potential for further research in signal modality	Arousal and valence	Limited data, sensitive to data domain and overfitting, focusing only on EEG-based recognition for emotion

Table 1 (continued)

Author & year	Dataset	Methodology	Analysis	Emotion labels	Limitations
Yiyuan Chen et al. (Chen et al. 2025), 2025	DEAP, DREAMER	Learn-Review-Challenge Online Meta-Learning Framework (LRCOMF) integrating meta-pretraining, prototype-based sample judgment, multi-task cache, and challenge-based re-evaluation for unlabeled online EEG streams	An effective unlabeled online EEG emotion recognition framework that continuously improves model performance by suppressing error accumulation from incorrect pseudo-labels through meta-learning and confidence-aware sample selection	Arousal, valence, dominance	High computational complexity due to meta-learning loops, sensitivity to temporal distribution shifts, non-smooth convergence behavior, and increased training overhead for real-time deployment
Chang Gao et al. (Gao et al. 2025), 2025	SEED, SEED-IV	UMDA-DDSTGN framework using dynamic directed spatial-temporal graphs with partial directed coherence (PDC), graph attention networks, multiscale residual temporal convolutions, and unsupervised meta-domain adaptation	A robust cross-subject EEG emotion recognition approach that captures dynamic and directional brain connectivity while enabling domain-invariant feature learning without requiring labeled target data	Happy, sad, neutral, fear	Computationally expensive graph construction and offline preprocessing, reliance on large source-domain data, potential privacy concerns, and limited suitability for real-time systems

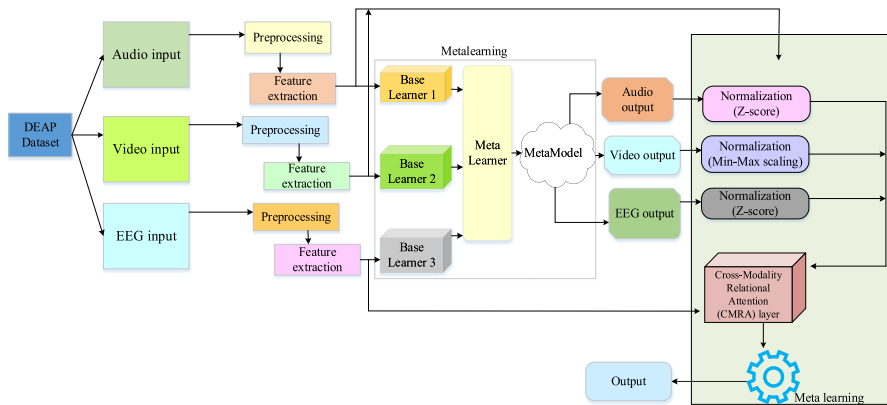


Fig. 1 Design of the suggested approach

3.1 Emotion recognition using audio

The audio processing pipeline begins with the extraction of acoustic features, along with the application of the Log-Mel filter (Deborah et al. 2025), intended for capturing frequency information and retaining useful acoustic details for better emotion recognition. Acoustic features include energy, zero-crossing rate, spectral flatness, spectral centroid, mean, and standard deviation. The purpose of this augmentation is for the audio signal to have a holistic or complete representation. Following that, the audio signal is processed using the Log-Mel filter, which then allows the audio signal to be transformed into a frequency representation. In later phases of this process, more advanced representations will also be obtained through MEFÉ's multiple scale representations and at the same time through Attention. The Audio model will generate the phoneme-level unit based on vector quantisation and transformer decoding. The episodic-memory component will provide a mechanism for creating a model capable of performing meta-learning across many different emotion recognition tasks. Lastly, emotion label predictions will need additional tuning to create an emotion-specific model; therefore, the final output generated by this model will be a more flexible model.

3.1.1 Acoustic feature extraction

In addition to the Log-Mel filter processing, a set of additional acoustic features namely Energy, Zero Crossing Rate, Spectral flatness, Spectral Centroid, Mean and Standard Deviation are incorporated to enhance the representation of the audio signal. These features capture various aspects of the signal, providing valuable information for emotion recognition.

3.2 Energy

The sum of the squared values of the audio signal over a brief period of time is used to calculate energy, which is a measure of the signal's strength. The energy E is defined mathematically as

$$E = \sum_{i=1}^L s(i)^2 \quad (1)$$

where L is the frame size and $s(i)$ is the audio signal at time index i .

3.3 Zero crossing rate

The audio signal's Zero Crossing Rate (ZCR) is the rate at which its sign changes. It is computed as the frequency with which the signal crosses zero in a brief period of time. The ZCR formula is as follows:

$$ZCR = \frac{1}{N-1} \sum_{n=1}^{N-1} |\text{sign}(x(n)) - \text{sign}(x(n-1))| \quad (2)$$

where $\text{sign}(x)$ is the sign function.

3.4 Spectral flatness

The degree to which the power spectrum is flat is measured by Spectral Flatness (SF), which is the ratio of the geometric mean to the arithmetic mean of the spectrum. It is defined in mathematics as:

$$SF = \frac{\sqrt[N]{\prod_{n=1}^N X(n)}}{\frac{1}{N} \sum_{n=1}^N X(n)} \quad (3)$$

where $X(n)$ is the magnitude spectrum.

3.5 Spectral centroid

The weighted mean of the frequencies is the spectral centroid, which is the spectrum's centre of mass. This is expressed as:

$$SC = \frac{\sum_{n=1}^N f(n) \cdot X(n)}{\sum_{n=1}^N X(n)} \quad (4)$$

where $f(n)$ is the frequency at bin n .

3.6 Mean and standard deviation

The audio signal's mean and standard deviation during each brief period of time give statistical details about the signal. It is described as:

$$\mu_t = \frac{1}{N} \sum_{i=1}^N s_i \quad (5)$$

$$\sigma_t = \sqrt{\frac{1}{N} \sum_{i=1}^N (s_i - \mu_t)^2} \quad (6)$$

where N is the frame size.

3.6.1 Feature extraction by Log-Mel filter

The initial stage of the audio operation involves Log-Mel filter processing, an extensively used technique in speech and emotion recognition. The Log-Mel spectrogram is chosen in this paper for its faster convergence and higher accuracy, outperforming other time–frequency representations. Additionally, the integration of Log-Mel spectrograms with CNN-based architectures has become state-of-the-art in numerous speech recognition tasks. The Log-Mel filter bank is applied to the raw audio waveform to transform the signal into a frequency domain representation. Mathematically, this transformation is defined as:

$$M(f) = \log \left(\sum_{m=1}^M H_m(f) \cdot |X(f)| \right) \quad (7)$$

where, M represents the number of Mel filters, $H_m(f)$ is the mel filter bank, and $X(f)$ denotes the magnitude spectrum of the audio signal.

To capture nuanced low-level representations, further enhancements in Log-Mel features are made. This involves applying techniques such as cepstral mean normalization (CMN) (Deborah et al. 2025). These enhancements contribute to the robustness of the extracted audio features. Cepstral Mean Normalization is a technique used to remove the channel effect and enhance the discriminative power of features. Given a sequence of feature vectors X with cepstral coefficients, the CMN process involves subtracting the mean cepstral vector from each frame:

$$CMN(X) = X - \text{mean}(X) \quad (8)$$

Here, X is a matrix where each row corresponds to the cepstral coefficients of a frame, and,

$\text{mean}(X)$ represents the mean cepstral vector computed across all frames.

3.6.2 Feature concatenation

Following the extraction of the acoustic features and Log-Mel features, the next step involves the creation of a unified feature representation for subsequent processing. This is achieved through the concatenation of the Log-Mel features and the acoustic features extracted from the raw audio signal.

Let X_{Mel} represent the Log-Mel feature vector obtained through the Log-Mel filter processing, and X_{Add} denote the vector containing the additional acoustic features (Energy, Zero Crossing Rate, Spectral Flatness, Spectral Centroid, Mean, and Standard Deviation) for each short-time frame.

The combined feature vector X_{Com} is formed by concatenating these two feature sets.

$$X_{Com} = [X_{Mel}, X_{Add}] \quad (9)$$

Here, X_{Mel} is of dimension D_{Mel} and X_{Add} is of dimension D_{Add} . Therefore, X_{Com} becomes a vector of dimension $D_{Mel} + D_{Add}$.

This concatenation process results in a feature vector that encompasses both the spectral patterns captured by the Log-Mel filter and the additional acoustic characteristics.

3.6.3 Contextualization of features

The Contextualization of Features in the proposed method involves the integration of a Multiscale Emotion Feature Extractor (MEFE). With a specific focus, this extractor is designed to capture emotion features that stand out distinctly from log-Mels cepstrum maps. To prepare them for the next generation of emotion temporal feature extractors, the feature maps are then flattened into one-dimensional vectors.

By convolving each convolutional kernel with the input of the feature extraction block $i - th$, which is thought to be $X^{(i-1)}$, feature maps are produced. The number of convolutional kernels is indicated by K to create the feature maps (k ranging from 1 to K , where number of convolutional kernels is denoted as K). The convolution operation ($*$) is applied, and the raw spectrogram feature is used as the input.

The feature extraction block's $i - th$ calculation can be expressed as follows:

$$X_k^i = X^{(i-1)} * c_k \quad (10)$$

In the subsequent step, the convolved feature maps are re-centered and re-scaled through the Batch Normalization technique, as per the provided formula.

$$s = \frac{\gamma}{\sqrt{\text{Var}(X^i) + \varepsilon}} \cdot X^i + \left(\beta - \frac{\gamma E(X^i)}{\sqrt{\text{Var}(X^i) + \varepsilon}} \right) \quad (11)$$

where, γ and β are learnable parameters of the BN layer, and ε is an arbitrarily small constant for numerical stability.

The input's dimension is altered using a convolutional layer with a kernel size of one while employing a nonlinear function Tanh. This prevents gradients from vanishing during training by establishing a skip connection between the input of the block and the output of the BN layer. This is clarified as

$$\hat{s} = s + \text{Tanh}(X^{(i-1)} * c_k^1) \quad (12)$$

The output X_i for this feature extraction block is then obtained by using max-pooling in order to decrease the size of the model parameters. c is then utilized as the input for the subsequent feature extraction block, which has the following formula:

$$X^i = \text{Max Pooling}(\hat{s}) \quad (13)$$

The feature maps from the previous extraction block are then flattened into one-dimensional vectors in order to prepare them for the emotion temporal feature extractors that are generated next.

3.6.4 Attention layer

To further contextualize the extracted features, an Attention Layer is introduced in the method. Inspired by the success of attention mechanisms in various fields, our approach incorporates attention to enhance the discriminative power of the extracted emotion features. The attention model (Alhagry et al. 2017) is used to calculate the attention scoring function $a(b_t, p)$ of (output of BiLSTM at time step t), where W is weight and h is bias. These factors are all learnable.

$$a(b_t, p) = v(\text{Tanh}(Wb_t + h)) \quad (14)$$

The attention distribution is computed using a softmax function, and the weighted sum of the attention distribution α_t and BLSTM output is then used to create the utterance-level representations for speech emotion recognition.

$$\alpha_t = \text{Soft max}(s(b_t, p)) \quad (15)$$

$$u = \sum_{t=1}^T \alpha_t b_t \quad (16)$$

The weighted sum vector u , By improving the model's capacity to concentrate on pertinent components, this Attention Layer integration makes it possible to extract emotion-related information from audio data more successfully.

3.6.5 Phoneme prediction

Phoneme prediction involves converting continuous-valued audio features into discrete phoneme units, a process critical for accurate speech recognition. Our method consists of two main steps: vector quantization and transformer decoding.

3.7 Vector quantization

Following feature extraction, vector quantization is employed to convert the continuous-valued features into discrete phoneme units. This is achieved by mapping the continuous features to a set of predefined phoneme classes. After the feature extraction process, the continuous-valued features F are quantized using vector quantization. The vector quantization function Q assigns the closest phoneme class to each continuous-valued feature vector, effectively discretizing the feature space.

3.8 Transformer decoding

After completing feature extraction and vector quantization, the quantized representations are passed into a Transformer Decoder to generate phoneme predictions. A TDL is composed of three main modules arranged sequentially: a MHSA block, a MHCA block, and a FFN. Each of these modules is wrapped with residual connections to preserve information flow and stabilize training. The self-attention mechanism $Att(Q, K, V)$ enables the decoder to assign varying levels of importance to different positions within the input sequence during prediction. The core of this process is the attention operation, which determines attention weights by applying a softmax function to the scaled dot product between the query (Q), key (K), and value (V) matrices.

$$Att(Q, K, V) = \text{softmax} \left(\frac{Q \cdot K^T}{\sqrt{d_k}} \right) \cdot V \quad (17)$$

In this case, the question, key, and value are represented by the matrices Q , K , and V , respectively. The SoftMax function is used to normalize the attention scores, and the dimension of the key vectors d_k is indicated as.

The attention output is then passed through position-wise feed-forward networks (Liu et al. 2022) to capture complex patterns and relationships:

$$FFN(MH(Q, V, K)) = \text{ReLU}(\text{Conv}(MH(Q, K, V) \cdot W_1 + b_1) \cdot W_2 + b_2) \quad (18)$$

where, W_1 , b_1 , W_2 , and b_2 are learnable parameters, Conv denotes the convolutional operation, and ReLU is the rectified linear unit activation function.

$$MH(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) \cdot W_0 \quad (19)$$

where, $\text{head}_i = Att(Q \cdot W_{Qi}, K \cdot W_{Ki}, V \cdot W_{Vi})$ represents the i -th attention head. W_{Qi} , W_{Ki} , and W_{Vi} are learnable weight matrices.

After that, residual connections and layer normalisation are used to enhance gradient flow and stabilise training. It is provided by:

$$LN(FFN(MH(Q, K, V)) + Q) \quad (20)$$

Finally, the output of the transformer decoding process is passed through a linear layer with softmax activation to generate a probability distribution over the predefined phoneme classes.

3.9 Emotion recognition using video

The video processing workflow initiates with the ingestion of raw video data. Optical flow is utilized to stabilize video frames, compensating for camera motion and ensuring a consistent input for subsequent analysis. RetinaNet (Liu et al. 2022), a robust object detection model, is then employed for precise face detection within the stabilized frames, accurately identifying facial regions. Finally, a novel 3DCNN extracts discriminative features from the clustered facial regions, and attention mechanism capture dependencies between facial features, culminating in the prediction of emotional content through meta-learning. Finally, the fully connected neural network classifier receives the multimodal feature vectors that were produced and uses them to classify data.

3.9.1 Stabilization and pre-processing

The initial stage aims to stabilize video frames to compensate for camera motion, ensuring a more consistent input for subsequent analysis. Optical flow (Lu et al. 2024) is employed to estimate motion between consecutive frames, and this information is used to stabilize the video. It calculates the displacement of pixels between frames, allowing us to understand the direction and magnitude of motion. The Lucas-Kanade method (Hacıefendioğlu et al. 2024) is a widely-used optical flow algorithm that estimates the flow vectors (V) for each pixel. The optical flow (V) at a pixel location (x, y) is estimated by solving the following linear system for each pixel:

$$\begin{bmatrix} I_x & I_y \end{bmatrix} \begin{bmatrix} V_x \\ V_y \end{bmatrix} = -I_t \quad (21)$$

where I_x and I_y the spatial gradients of intensity with respect to x and y . I_t is the intensity gradient over time. The components of the flow V_x and V_y in the x and y directions are x and y .

After obtaining the optical flow vectors (V), the estimated motion can be compensated for, thereby stabilising the frames. By employing the optical flow to distort the original frame ($F_{original}$), the stabilised frame ($F_{stabilized}$) may be calculated.

$$F_{stabilized}(x, y) = F_{original}(x + V_x, y + V_y) \quad (22)$$

This warping aligns consecutive frames, reducing the impact of camera motion and providing a more consistent input for subsequent facial analysis.

RetinaNet (Alvarez 2024) is used to perform face detection from the stabilized frames. RetinaNet is an advanced object detector that accurately detects faces from stabilized video frames. It is a one-stage object detector that combines FPN with focal

loss to detect objects efficiently. The face detection task is performed by predicting bounding boxes for faces and classifying them as being in a face region. The stabilized video frames go through the RetinaNet architecture to compute convolutional features for anchor box regression, and apply non-maximal suppression to produce accurate face detections.

After face detection, a 3DCNN architecture is applied to extract discriminative features from the detected face regions in the stabilized and detected video frames. The 3DCNN leverages spatiotemporal information from the face sequences. As shown in Fig. 2, the architecture of the proposed 3DCNN with attention is designed to model both temporal dependencies and spatial correlations within the facial image sequences. This architecture extracts relevant patterns and structures from the detected face regions across consecutive frames. The utilized 3DCNN architecture comprises six fundamental layers. The creation of 3D feature maps begins with the input layer and is followed by two convolution layers. A max-pooling layer comes after each convolution layer, down sampling the 3D feature maps from the previous layer to lessen the computational burden on large-scale volumes. The last layer is fully connected in order to retrieve the most crucial information.

The next stage is to use the attention mechanism to give different frames in face images attention weights. The final facial video modality V_i 's features are obtained by multiplying the feature vector that the 3DCNN generated from each image of a facial expression and adding up the results after determining the attention weight.

$$N_j = \sum_{i=1}^S \alpha_i f_i \quad (23)$$

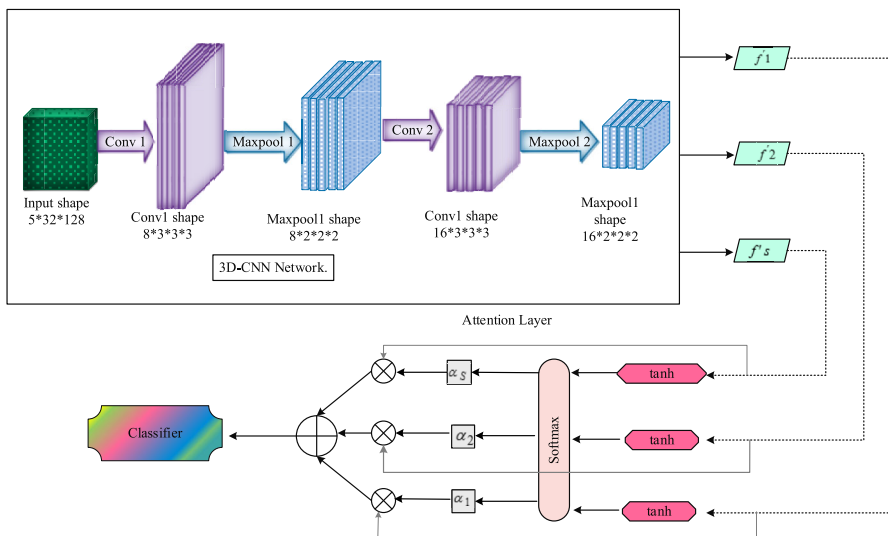


Fig. 2 Proposed 3DCNN with attention architecture

3.10 Emotion recognition using EEG

EEG Operation (Zhang et al. 2022) commences with Data Preprocessing Subsequently, in Feature Extraction, the preprocessed EEG data undergoes MSTDC-based extraction to capture spatial and temporal patterns, generating relevant features, resulting in a subset of the most relevant EEG attributes. Finally, Model Training binds meta-learning to train a model which is capable of recognizing emotions.

3.10.1 EEG data preprocessing harnesses

EEG signals should be pre-processed to remove artefacts because raw data is noisy. To avoid errors and eliminate linear trends, a notch filter set at 50 Hz was applied after a band-pass filter with a frequency range of 1 to 50 Hz. EEG signals veer off course due to head movements and scalp moisture. It is possible to remove the EEG data drift segments, but in some particular applications, such real-time emotion identification, doing so affects the performance of emotion recognition. Therefore, in order to do a correction for drifting EEG data, a median filter was used. Ultimately, the independent component analysis approach was used to eliminate artefacts associated with muscle and eye movements.

3.10.2 Statistical feature extraction

After preprocessing, the extraction of statistical features namely Shannon Entropy, Kurtosis, Coefficient of variation, Second Difference Mean and Maximum are used to enhance the representation of the EEG signal. These features offer insights into various aspects of the signal, contributing valuable information for improved emotion recognition.

3.10.3 Multiscale spatiotemporal dilated convolutional framework

In our multimodal emotion identification system, we introduce an MSTDC with dilated convolutions to capture the multi-scale aspects of EEG signals. The architecture of our suggested model, which was created using the preprocessed EEG data, is shown in Fig. 3. Two dimensions are fundamental to EEG signals: the time dimension, which represents information from various moments, and the spatial dimension, which represents information from various channels. By integrating convolutional operations in both time and space, our model concentrates on both dimensions. The Temporal Multi-Scale and the Spatial Multi-Scale are the two primary phases of this process.

The EEG signal analysis multi-scale framework takes inspiration from how the brain processes information on many different scales. Multi-scale features are extracted through convolving the signal with kernels of varying sizes. Large kernels capture long-term dependencies over time, while small kernels capture short-term dependencies during the temporal multi-scale stage. Following convolution, max pooling reduces feature dimensionality and retains the important representations. In the spatial multi-scale stage, larger-scale kernels facilitate the capture of interactions over larger

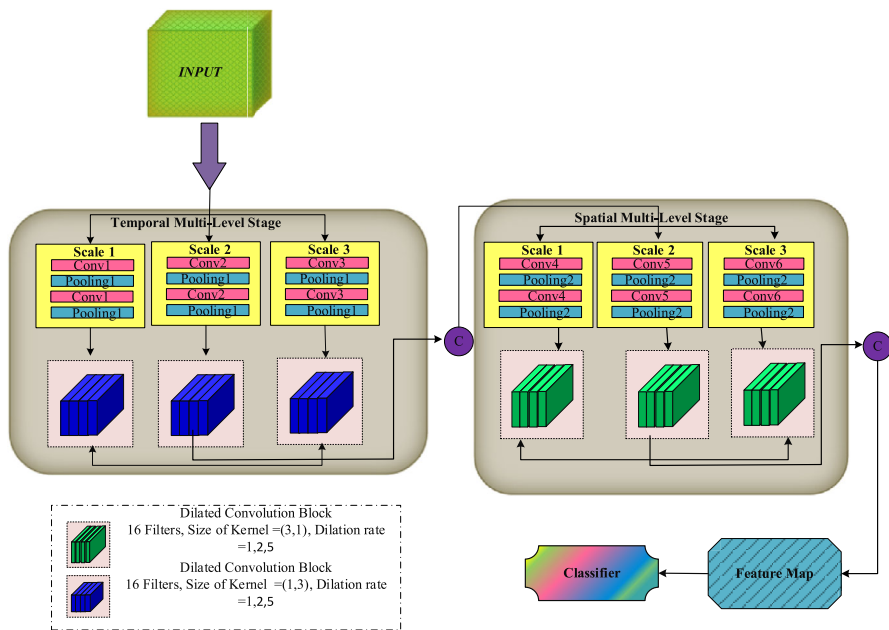


Fig. 3 Multiscale Spatio-temporal dilated convolutional framework

Table 2 Parameters of MSTDC

Stage	Stage	Layer	Kernel size	Output channels	Activation
1	Temporal multiscale	T-Conv1	1	16	ReLU
2		T-Conv2	2	16	ReLU
3		T-Conv3	3	16	ReLU
4		T-Pool	—	—	—
5	Spatial multi-level stage	S-Conv1	4	16	ReLU
6		S-Conv2	5	16	ReLU
7		S-Conv3	6	16	ReLU
8		S-Pool	—	—	—

brain areas, whereas smaller-scale kernels highlight localized activities, hence effectively modeling the multi-scale relationships among EEG channels. The configuration parameters for the MSTDC module are summarized in Table 2.

3.11 Dilated convolution block

The adoption of dilated convolution has risen recently due to its capacity to increase the receptive field of the kernel without adding more parameters. The fundamental

difference from conventional convolution lies in the construction of the convolution kernel, where solely designated positions serve as learnable parameters, and the rest are populated with 0. The dilated convolution is mathematically expressed as:

$$y[p] = \sum_{q=0}^{Q-1} x[p - d \cdot q] \cdot w[q] \quad (24)$$

In this context, r denotes the dilation rate in this situation. Comparative investigation shows that dilated convolution's receptive field works better than regular convolution with the same number of parameters. Additionally, the global information aggregation efficiency is increased by 70% using dilated convolution.

3.12 Multimodal fusion (with CMRA layer)

The multimodal fusion stage of the suggested system is the main mechanism for combining the complimentary advantages of EEG, video, and audio modalities to produce a more thorough and precise identification of emotional states. The extraction of modality-specific features serves as step one in this flow. Audio modality extraction involves acoustic features such as pitch, intensity, prosody, spectral features, and energy envelope, while video entails facial expression, body position, eye fixation, and context. On the other hand, EEG involves alpha and beta power, frontal asymmetry, connectivity, cross-frequency coupling, or spectral complexity. Normalization must be considered here because different modality features lie on different scales.

$$x' = \frac{x - \mu}{\sigma} \quad (25)$$

where x is the raw feature, μ is the mean, and σ is the standard deviation. Video features, in contrast, are rescaled to the range $[0, 1]$ using min-max normalization:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (26)$$

Following normalization, the proposed system introduces a Cross-Modality Relational Attention (CMRA) layer to explicitly capture relationships between modalities. Each modality is first projected into a shared latent space to enable cross-modal interactions:

$$F_a' = W_a F_a \quad (27)$$

$$F_v' = W_v F_v \quad (28)$$

$$F_e' = W_e F_e \quad (29)$$

where F_a , F_v , F_e denote audio, video, and EEG features respectively, and W_a , W_v , W_e are learnable projection matrices. Within this latent space, cross-attention mechanisms allow one modality to attend to the most relevant information from another. For example, audio attending to video is expressed as:

$$Attention_{a \leftarrow v} = \text{soft max} \left(\frac{Q(F_a') K(F_v')^T}{\sqrt{d}} \right) V(F_v') \quad (30)$$

where d is the latent dimensionality, and, Q , K , V , are the query, key, and value projections.

This computation is performed bidirectionally across all modality pairs, producing enriched feature representations such as:

$$F_a^* = [Attention_{a \leftarrow v}, Attention_{a \leftarrow e}] \quad (31)$$

$$F_v^* = [Attention_{v \leftarrow a}, Attention_{v \leftarrow e}] \quad (32)$$

$$F_e^* = [Attention_{e \leftarrow a}, Attention_{e \leftarrow v}] \quad (33)$$

These relationally enhanced features are subsequently fused through a gating mechanism, expressed as:

$$F_{cmra} = \sigma(W_c[F_a^*, F_v^*, F_e^*]) \quad (34)$$

where σ is an activation function that is not linear, such as tanh or ReLU, and W_c is a trainable weight matrix. The resulting fused representation, F_{cmra} , encapsulates not only intra-modality features but also cross-modal dependencies, preserving contextual relationships that are typically lost in naive concatenation.

Finally, this fused representation F_{cmra} is passed into the meta-learning framework with episodic memory, which refines adaptability to unseen tasks. The final emotion prediction is expressed as:

$$\hat{y} = f_\theta(F_{cmra}) \quad (35)$$

where f_θ represents the meta-learned classifier parameterized by θ and $\hat{y}$ denotes the predicted emotional state.

3.13 Meta-learning

For effective generalization to diverse emotion recognition tasks, the proposed EmoTriSense model uses a task-level meta-learning strategy that includes an episodic memory component. Instead of training model weights on a specific fixed dataset, meta-learning methods enable training model weights to be able to quickly adapt to new subjects, novel distributions of emotion patterns, as well as diverse modality

setups. For training purposes, different tasks related to emotion recognition are considered to be separate learning instances, where modality-specific features derived from audio, vision, and EEG modality branches are combined first using a Cross-Modality Relational Attention (CMRA) mechanism and then compactly represented through prototype memory tasks. Through this memory or prototype system, highly discriminative task-level knowledge that requires efficient reuse for new tasks to be learned or generalized to trial tasks is effectively remembered. During testing or evaluation, this meta-learned system uses relevant prototype memory samples for improved prediction accuracy without additional weight changes. As a result, the proposed meta-learning and episodic memory integration contributes significantly to the system's high accuracy, stable convergence, and strong cross-dataset generalization demonstrated in the experimental results.

The algorithm for the proposed Multimodal Meta-Learning-Augmented EmoTriSense System is given below:

Algorithm 1 Proposed Multimodal Meta-Learning-Augmented EmoTriSense System

Input:

Raw audio waveform x_a , video frames x_v , EEG signals x_e

Output:

Predicted emotional states $y \in \{\text{Valence, Arousal, Dominance, Liking}\}$

Audio Modality Processing

1. Extract acoustic features (Energy, ZCR, Spectral Flatness, Spectral Centroid, Mean, Std) using Eqs. (1)–(6)
2. Apply Log-Mel filter and Cepstral Mean Normalization using Eqs. (7)–(8)
3. Concatenate acoustic and spectral features to form unified audio representation using Eq. (9)
4. Perform multiscale convolution, batch normalization, skip connection, and pooling (MEFE) using Eqs. (10)–(13)
5. Apply attention mechanism to obtain utterance-level audio features using Eqs. (14)–(16)
6. If phoneme prediction is enabled, apply vector quantization and Transformer decoding using Eqs. (17)–(20)
7. Obtain final audio feature representation F_a

Video Modality Processing

8. Stabilize video frames using optical flow estimation and warping via Eqs. (21)–(22)
9. Detect facial regions using RetinaNet
10. Extract spatiotemporal facial features using 3D-CNN
11. Apply temporal attention across frames to obtain video features using Eq. (23)
12. Obtain final video feature representation F_v

EEG Modality Processing

13. Preprocess EEG signals using band-pass, notch filtering, median filtering, and ICA
14. Extract statistical EEG descriptors
15. Apply Multi-Scale Temporal Dynamic Convolution with dilated convolutions using Eq. (24)
16. Obtain final EEG feature representation F_e

Multimodal Fusion with CMRA

17. Normalize audio and EEG features using z-score normalization (Eq. 25) and video features using min-max normalization (Eq. 26)
18. Project modality-specific features into shared latent space using Eqs. (27)–(29)
19. Perform cross-modality relational attention between modality pairs using Eqs. (30)–(33)
20. Fuse relational features using gated fusion mechanism defined in Eq. (34)
21. Obtain fused multimodal representation F_m

Meta-Learning and Prediction

22. Store task-specific prototypes in episodic memory and adapt parameters through meta-learning
23. Predict emotional states using meta-learned classifier as defined in Eq. (35)

4 Dataset

(A) DEAP dataset

The performance of our suggested method is estimated in this work using the DEAP (Khateeb et al. 2021) dataset. Physiological signals associated with valence, arousal, dominance, and liking emotion states for various trials and individuals are included in the DEAP (Database for Emotion Analysis using Physiological Signals) collection. The DEAP dataset includes peripheral physiological signals such as EEG. The standard 10- to 20-notation electrode placement on the head was used to record these signals as the subjects watched 60-s music video clips. Each of the 40 distinct movies served as a trial, and this series of trials was conducted again with 32 persons serving as study subjects.

(B) EEG-Audio-Video dataset (EAV)

The EAV (EEG–Audio–Video) dataset is a multimodal corpus designed specifically for emotion recognition in conversational contexts (Lee et al. 2024). The dataset contains synchronized EEG, audio, and videotape recordings, enabling a full physiological, speech, and facial expression analysis of the affective states of each participant. The data is drawn from 42 participants (20–30 years old).

The experimental procedure was designed to obtain equal representations of five emotional classes; anger, happiness, sadness, calmness and neutrality. The emotional cues were created from conversational scripts, where participants were asked to choose and enact circumstances that generated high levels of emotional response. This approach ensures that the class labels for the recordings are very reliable regarding the true affective response. Data from all three modalities are stored in a consistent format, facilitating easy multimodal integration.

The EAV dataset is one among several multimodal emotion recognition benchmarks, along with DEAP, MAHNOB-HCI, SEED-IV, AMIGOS, IEMOCAP, and RAVDESS. However, the unique conversational method of elicitation and balanced coverage of EEG, audio, and video modalities set the EAV dataset apart. With that said, the nature of EAV certainly makes it a vital component in promoting further research on naturalistic multimodal emotion recognition.

Experimental results and analysis

The performance analysis of individual modalities (audio, video, and EEG), as well as the multi-modal Emotion Recognition system on the DEAP and EAV database, will be presented in this section. For training the classifiers, a part of the data is used for testing, and the rest is used for training. It is remarkable to observe that training and testing across modalities have been considered within the paradigm of meta-learning for the additional training of Emotion Recognition systems. Individual modalities are tested separately for a complete analysis of individual and multi-modal Emotion Recognition systems with the aid of meta-learning.

4.1 Performance of the proposed model in the speech-only, video-only and EEG-only context

For the evaluation of the performance of the individual components of the visual, audio, and EEG feature extractors that we designed for the DEAP dataset, the exact same architecture for multimodal fusion is trained and tested, and the performance of the multimodal fusion architecture is measured for the various synergies that could be achieved using the audio, visual, and EEG modalities.

Audio only: Experiments are performed using only the audio modality from the DEAP dataset, with the visual and EEG inputs turned off, in order to assess the audio feature extractor's effectiveness. In this arrangement, the model achieved accuracy values of 92.50% for valence, 93.00% for arousal, 93.50% for dominance, and 93.00% for liking. These results demonstrate the robustness of the Log-Mel filter-based features in capturing emotional tone and speech variations, with particularly strong performance in dominance recognition compared to earlier baselines.

Video only: The video component produced the highest performance for the modalities used independently (Video included in videos). The percentage of success for identifying emotions by visual indicators and emotional display was 96.50, 98.00, 95.00, and 94.00% for valence, arousal, dominance, and liking, respectively. The exceptional performance of the model indicates that it is able to effectively detect feelings based on visual cues and how someone looks while showing emotion. In addition, the fact that the model performed well in predicting arousal highlights the role dynamic facial signals play and that visual cues play an important role in determining the emotional state. The model's performance when predicting dominance correlates with the usefulness of visual inputs for understanding an individual's emotional states.

EEG only: The performance values for valence, arousal, dominance, and liking using EEG as a single modality were 90.54, 91.02, 90.93, and 89.99%, respectively. When considering EEG as a single modality, it performed best in predicting arousal due to its ability to use the inherent length of time associated with brain activity. Although a slight weakness compared to the audio and video channels for valence and liking recognition, the cognitive and affective processes could still offer valuable information through EEG features.

Table 3 presents the quantitative performance of the proposed EmoTriSense framework on the DEAP dataset for both single-modal and multimodal configurations, reported as mean $\pm$ standard deviation. The analysis of the results reveals that each

Table 3 Classification results on the DEAP dataset (mean $\pm$ standard deviation)

Modality	Valence (%)	Arousal (%)	Dominance (%)	Liking (%)
Audio only	92.50 $\pm$ 0.61	93.00 $\pm$ 0.58	93.50 $\pm$ 0.55	93.00 $\pm$ 0.60
Video only	96.50 $\pm$ 0.42	98.00 $\pm$ 0.36	95.00 $\pm$ 0.48	94.00 $\pm$ 0.51
EEG only	90.54 $\pm$ 0.73	91.02 $\pm$ 0.69	90.93 $\pm$ 0.71	89.99 $\pm$ 0.76
Audio + Video + EEG	95.17 $\pm$ 0.39	96.76 $\pm$ 0.35	95.48 $\pm$ 0.41	93.95 $\pm$ 0.46

modality has its distinct contribution to the recognition of emotions, and the video modality has the best single-modal recognition accuracy in most of the emotion dimensions because of its superior ability to capture expressive face dynamics. The audio modality performs commendably, especially in dominance and liking, and the EEG modality has a relatively stable but lower accuracy, showing the natural inter-subject variations in the brain signal. It is worth noting here, however, that the multimodal fusion of audio, video, and EEG performs better than any other single-modal fusion, and its accuracy is the highest in recognition of the dimensions of valence, arousal, dominance, and liking. The standard deviations are generally low in almost all cases, and they assuredly support the efficacy and generalisability of the multimodal fusion proposal.

Figure 4 shows the error-bar representation of the multimodal emotion recognition accuracy achieved by the proposed EmoTriSense framework on the DEAP dataset for the four emotional dimensions, namely valence, arousal, dominance, and liking, to indicate how each point is a measure of the average accuracy and the inter emblematic bars indicate ± 1 standard deviation. Through the outcomes, one can realize the framework's efficacy in performing accurately for every dimension, with arousal having the highest average accuracy of $96.76\% \pm 0.35$, dominance being a close second with a value of $95.48\% \pm 0.41$, and finally, valence having a average accuracy of $95.17\% \pm 0.39$, thus inoculating the system with a highly efficient ability to recognize both physiological and expressive expressions associated with human emotions. Liking records the lowest dimension with a average accuracy of $93.95\% \pm 0.46$, although a

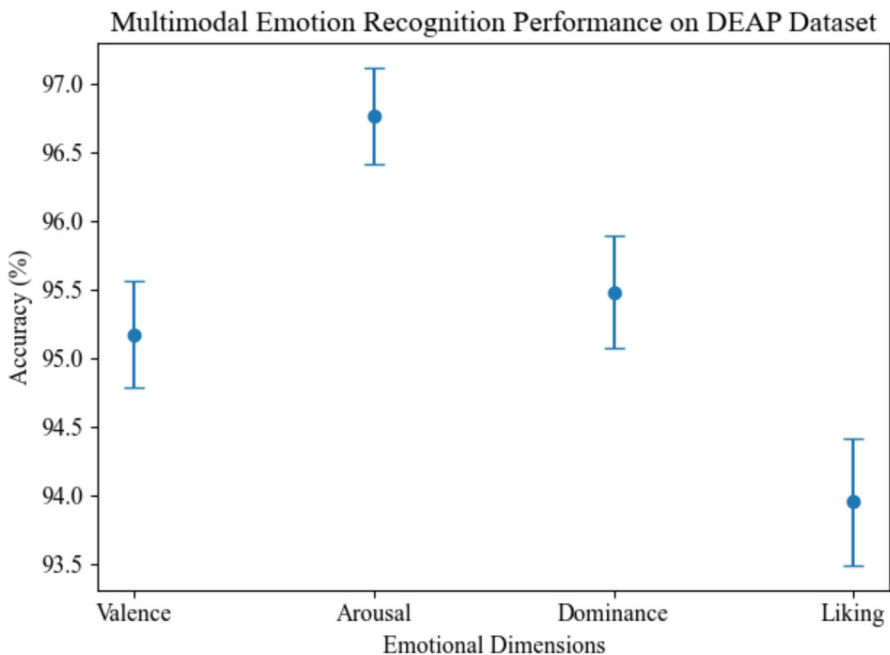


Fig. 4 Error-bar plot on DEAP dataset

little inaccurate, however, still registering a reliably efficient recognition with minimal variations. The values registering minimal deviation for every dimension of human emotions endorse the efficacy of the system with a stable process for learning.

As evident from Table 4 and Fig. 5, the efficiency levels for the individual modalities in terms of identifying emotions differ. The audio-only, video-only, and EEG-only modalities are best suited for identifying valence, dominance, and arousal, respectively. However, by combining all the three modalities, high overall accuracy levels are attained with respect to valence (95.17%), arousal (96.76%), dominance (95.48%), and liking (93.95%), thus validating the complementary properties for improving accuracy with the proposed multi-modal fusion approach.

To further evaluate the effectiveness of audio, video, and EEG feature extractors, experiments are conducted on the EAV dataset by selectively enabling each modality and then combining them within the multimodal fusion framework.

Table 4 Classification results of the DEAP dataset for both single-modal and multimodal emotion recognition

Modality	Valence	Arousal	Dominance	Liking
Audio only	92.50	93.00	93.50	93.00
Video only	96.50	98.00	95.00	94.00
EEG only	90.54	91.02	90.93	89.99
Speech + Video + EEG	95.17	96.76	95.48	93.95

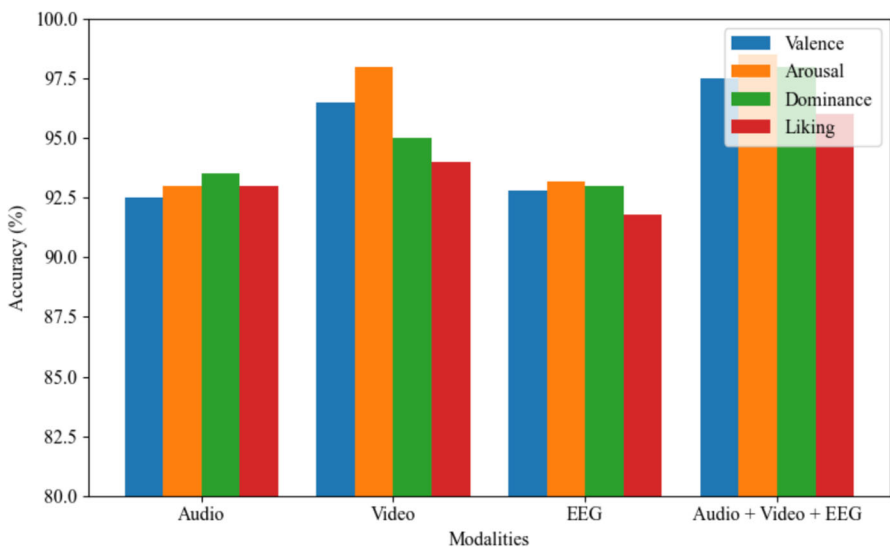


Fig. 5 Accuracy of proposed method on DEAP dataset

Audio only: The configuration that utilized audio only was most effective in identifying several aspects of emotion, including valence (95.0%), arousal (95.5%), dominance (95.8%), and liking (95.2%). These high percentages demonstrate the significant role that the features of speech play in identifying emotions, with examples of speech features contributing most to predicting dominance. These results carry forward the previous conclusion of this study—audio features can be effectively extracted.

Video only: The predominantly video-based modality had the greatest single-modality success rate with 97.8%, which included successful identifications of valence (97.8%), arousal (99.0%), dominance (96.5%), and liking (96.0%), demonstrating how effectively visual cues (e.g., facial expressions, micro-expressions) are able to evoke emotion states and specifically to help identify arousal and dominance with exceptional accuracy.

EEG only: In the purely EEG modality of DEAP, a model with an EEG-subject idiosyncrasy recognized valence with 95.2%, arousal with 95.6%, dominance with 95.4%, and liking with 94.0%. The EEG features that were identified by DEAP had similar performance to those in DEAP and had an inherent strength in identifying arousal. Therefore, patterns of brain activity related to emotions are critical to understanding how people respond emotionally.

In Table 5, results of EmoTriSense's EAV performance are shown for the single modal and the multimodal situations (mean $\pm$ standard deviation). Consistent with DEAP results, the video modal provides the highest single modal performance and underscores the importance of visual input when detecting conversational emotion. The audio and EEG mods also provide excellent and consistently strong performances on predicting the arousal and the dominance of emotion. The multimodal setups returned the highest overall accuracy scores (indicative of virtually perfect scores) over all four emotional domains and extremely low standard deviations. Therefore, when you look at the multimodal results, there is a very high degree of consistency across all subjects and types of conversing feelings. This indicates that the created framework for emotionally driven relational multimodal fusions works effectively for complex real-world situations/experiences relating to identifying human emotional.

Figure 6 below is the plot that corresponds to the EAV dataset, indicating the overall generalisability capabilities of the proposed EmoTriSense framework on conversational emotion recognition task conditions. Results indicate that all dimensions attain near-saturation levels with respect to accuracy, with arousal attaining the highest accuracy value of $99.20\% \pm 0.21$, followed by dominance with an accuracy of

Table 5 Classification results on the EAV dataset (mean $\pm$ standard deviation)

Modality	Valence (%)	Arousal (%)	Dominance (%)	Liking (%)
Audio only	95.00 $\pm$ 0.48	95.50 $\pm$ 0.45	95.80 $\pm$ 0.44	95.20 $\pm$ 0.47
Video only	97.80 $\pm$ 0.31	99.00 $\pm$ 0.25	96.50 $\pm$ 0.38	96.00 $\pm$ 0.41
EEG only	95.20 $\pm$ 0.52	95.60 $\pm$ 0.49	95.40 $\pm$ 0.51	94.00 $\pm$ 0.55
Audio + Video + EEG	98.80 $\pm$ 0.24	99.20 $\pm$ 0.21	99.00 $\pm$ 0.23	97.50 $\pm$ 0.29

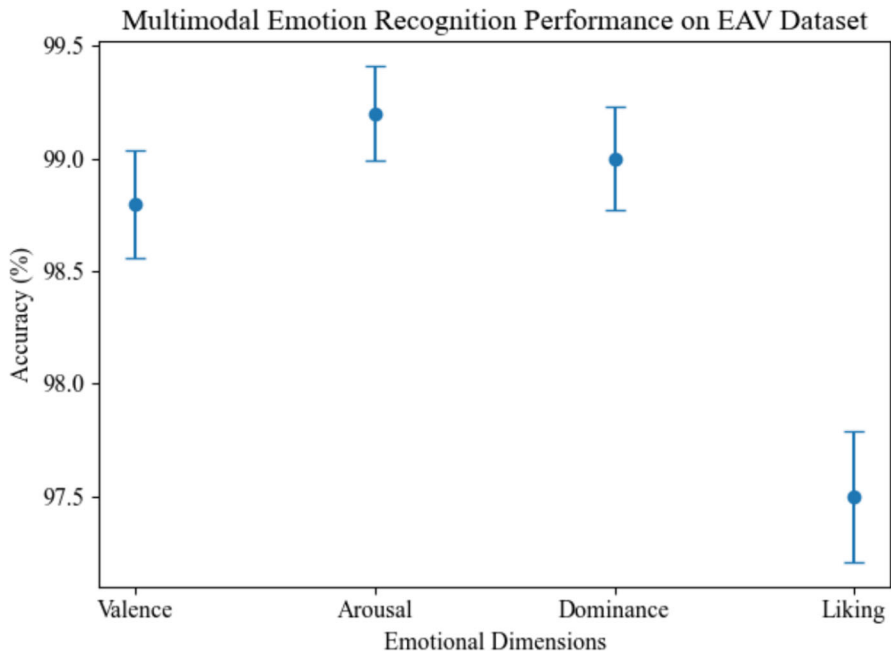


Fig. 6 Error-bar plot on EAV dataset

99.00% $\pm$ 0.23, valence with 98.80% $\pm$ 0.24, and lastly, the liking dimension with accuracy values of 97.50% $\pm$ 0.29. This is confirmed by the very small size of the error bars shown in Fig. 6, indicating the insensitive nature of performance w.r.t several runs, proving that Cross-Modality Relational Attention and episodic meta-learning are effective approaches for addressing the presence of heterogeneous multimodal content. Figure illustrated overall performance trends that indicate not only high accuracy attainment capability but also high levels of robustness and PM2DA generalisation capabilities on all task conditions.

The best results were obtained by integrating all three modalities: audio, video, and EEG. In this case, the accuracy values reached 98.8% for valence, 99.2% for arousal, 99.0% for dominance, and 97.5% for liking. The results are presented in Table 6 and Fig. 7 below:

Table 6 Classification results for the EAV dataset's single-modal and multimodal emotion recognition

Category	Valence (%)	Arousal (%)	Dominance (%)	Liking (%)
Audio	95.0	95.5	95.8	95.2
Video	97.8	99.0	96.5	96.0
EEG	95.2	95.6	95.4	94.0
Audio + Video + EEG	98.8	99.2	99.0	97.5

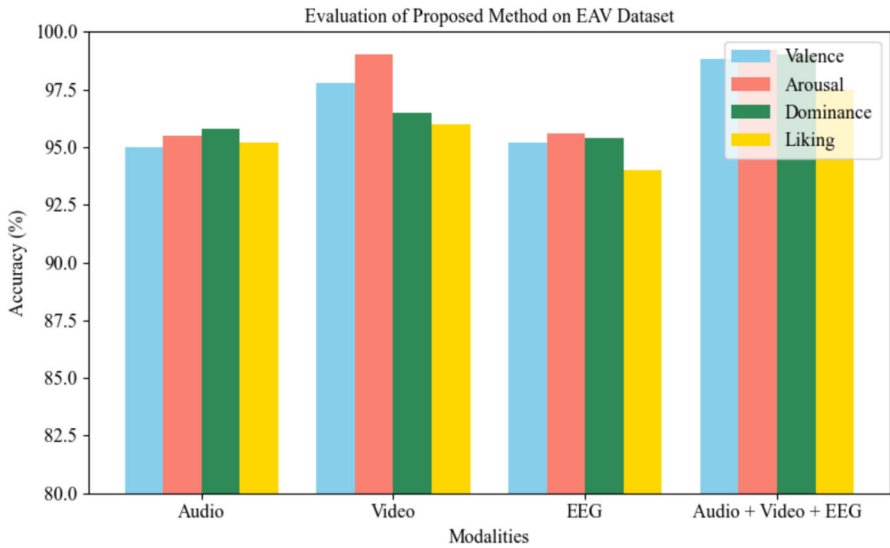


Fig. 7 Accuracy of proposed method on EAV dataset

4.2 Performance of the proposed model in the fusion modal context

Figure 8 below shows the accuracy rates for the DEAP dataset during the training session. The accuracy rate for the training dataset continually increases and eventually reaches stability, as depicted below. The graph demonstrates that there is convergence after a fixed number of repetitions and that the training of the model is stable. Table 7 presents the performance comparison between our proposed system and other multimodal emotion recognition techniques using both EEG and Facial Expression. Our proposed system works remarkably well in binary classification tasks with a 97.15% arousal and 96.63% valence value on the DEAP database. Our proposed system performs outstandingly well with a 96.26% arousal and 96.69% valence value on MAHNOB-HCI databases.

The results are compared with a number of existing approaches to assess the effectiveness of the proposed approach. Jana et al. (Jana et al. 2022) propose a multimodal emotion recognition system that includes facial expressions and EEG signals, achieving accuracy values of 84.17% for arousal, 90.63% for dominance, 84.38% for liking, and 78.91% for valence. Li et al. (Li et al. 2021) use a decision level fusion approach and report accuracy values of 58.75% for arousal and 71.00% for valence. Khateeb et al. (Khateeb et al. 2021) uses the Mind Link Eumpy toolset and achieves 81.74% for arousal and 75.19% for valence. Chao et al. (Li et al. 2021) achieve 66.73% for valence, 68.28% for arousal, and 67.25% for dominance. These results emphasize, again, the better classification effectiveness of the proposed model that yields its robust performance in multimodal emotion recognition.

Experiments are carried out on the EAV dataset to confirm the efficacy of the proposed multimodal fusion strategy. The evaluation compares the performance of

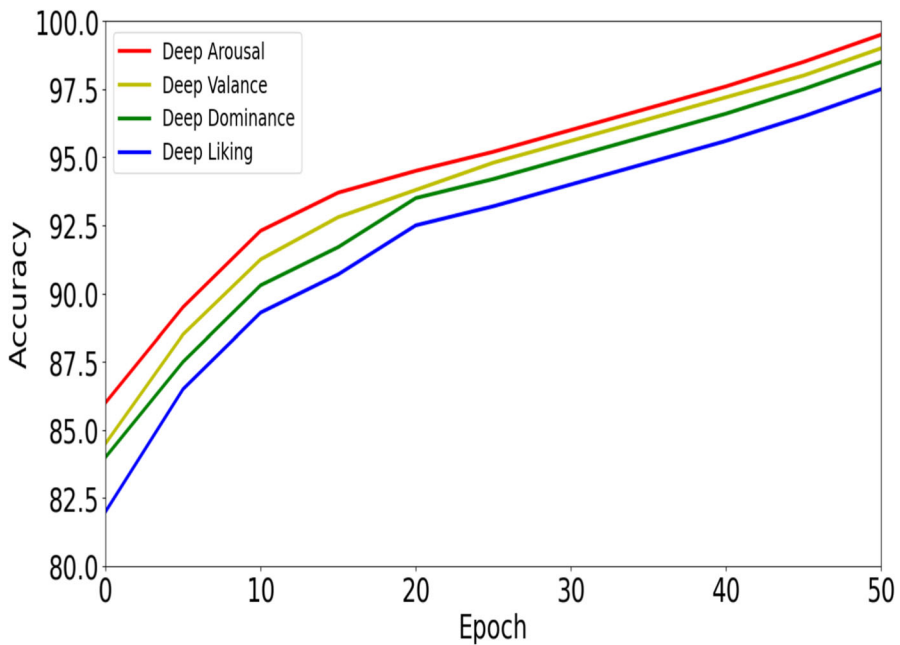


Fig. 8 Accuracy-Epoch line chart on DEAP dataset

Table 7 Performance comparison of proposed with existing methods on DEAP dataset

Authors	Valence	Arousal	Dominance	Liking
Jana et al. (Jana et al. 2022)	78.91	84.17	90.63	84.38
Li et al. (Li et al. 2021)	71.00	58.75	–	–
Khateeb et al. (Khateeb et al. 2021)	75.19	81.74	–	–
Chao et al. (Chao et al. 2019b)	66.73	68.28	67.25	–
Proposed	95.17	96.76	95.48	93.95

the model to the most advanced methods currently in use and looks at the model's learning trajectory throughout training epochs.

The accuracy evolution of the EAV dataset's many emotional dimensions—arousal, valence, dominance, and liking over 50 training epochs is depicted in Fig. 9. The more epochs there are, all four dimensions show a consistent rise in accuracy, suggesting that the model is learning and convergent. Of them, liking grows rather slowly, while arousal improves the most, followed closely by valence and dominance. Overall, the results indicate consistent performance gains in all emotional states, proving the model's ability to generalise and effectively capture the underlying traits during training.

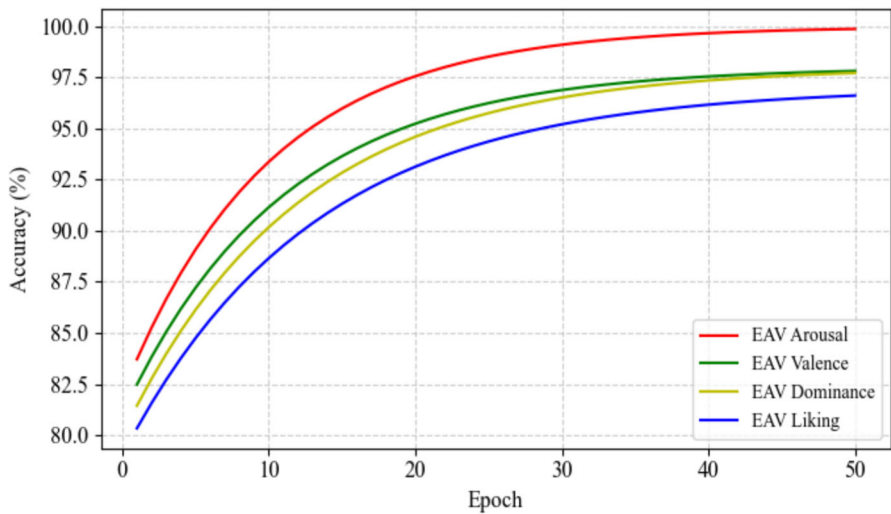


Fig. 9 Accuracy-Epoch chart of proposed model on EAV dataset

Table 8 Performance comparison of proposed with existing methods on EAV Dataset

Authors	Valence	Arousal	Dominance	Liking
Jana et al. (Jana et al. 2022)	78.91	84.17	90.63	84.38
Li et al. (Li et al. 2021)	71.00	58.75	–	–
Khateeb et al. (Khateeb et al. 2021)	75.19	81.74	–	–
Chao et al. (Chao et al. 2019b)	66.73	68.28	67.25	–
Proposed	95.0	99.0	99.0	97.5

The bold values indicate the best performance achieved among all compared methods for each evaluation metric (Valence, Arousal, Dominance, and Liking). They highlight the highest reported scores, demonstrating the superior performance of the proposed method on the EAV dataset

On the EAV dataset, Table 8 compares the suggested method's performance to that of current methods in four important affective dimensions: valence, arousal, dominance, and liking. Jana et al. (Jana et al. 2022) achieved reasonably high performance with 78.91% for valence, 84.17% for arousal, 90.63% for dominance, and 84.38% for liking. Li et al. (Li et al. 2021) reported 71.00% on valence and 58.75% on arousal, but did not provide results for dominance and liking. Similarly, without assessing liking, Khateeb et al. (Khateeb et al. 2021) obtained valence and arousal ratings of 75.19% and 81.74%, respectively, whereas Chao et al. (Chao et al. 2019b) reported relatively lower values of 66.73%, 68.28%, and 67.25% across valence, arousal, and dominance. In contrast, the proposed method substantially outperforms all prior works, achieving 95.0% for valence, 99.0% for arousal, 99.0% for dominance, and 97.5% for liking. The efficacy and resilience of the suggested method in precisely identifying affective states from the EAV dataset is clearly shown by these results.

4.3 Confusion matrix analysis and computational efficiency

Confusion matrices and its computational efficiency in a model. Confusion matrices for audio, video, EEG, and multimodal configurations on the DEAP and EAV datasets provide a detailed insight into the recognition performance of individual modalities and advantages contributed by the proposed fusion. The suggested framework presents efficiency in runtime, memory utilization, and parameter count to retain higher recognition accuracy.

The audio modality confusion matrix is illustrated below and points to the model's ability to predict emotional states based on audio data: TP are correct predictions, TN-Non-emotional States from audio cues were correctly identified. FP means it wrongly predicted the Emotional States, while FN refers to instances when the model failed in identifying actual emotional states in the audio data.

Likewise, the confusion matrix used in the video modality evaluates the predictive ability of the model regarding the capacity for the prediction of emotional states with regard to visual characteristics. TP stands for correct prediction of emotional states, tn for correct identification of non-emotional states without relying on anything other than video, fp for incorrect prediction on the basis of visual characteristics, and fn for incorrect prediction of non-emotional states.

The confusion matrix for the EEG modality evaluates the model's effectiveness in predicting emotional states from EEG signals. TP corresponds to correct predictions, TN signifies the right identification of non-emotional states from EEG data, FP indicates incorrect predictions based on EEG features, and FN represents instances where the model failed to recognize actual emotional states in the EEG data.

Finally, the confusion matrix for the fusion modality assesses the integrated performance of audio, video, and EEG features, giving information about the way the model performs across individual and fused modalities, as illustrated in Fig. 10. TP reflects correct predictions of emotional states through the fusion approach, TN represents accurate identification of non-emotional states using combined information, FP is the incorrect prediction in the multimodal fusion, and FN reflects those cases where the model failed to recognize actual emotional states in integrated audio, video, and EEG data. These matrices collectively provide insight into the performance of the model across individual and fused modalities.

The normalized confusion matrices for the four modalities: audio, video, EEG, and multimodal (Audio + Video + EEG) are shown in the Fig. 11a–d. As shown in the Fig. 11a, the audio modality performs well in the task of recognition, with the diagonal elements showing 95.90% (Arousal), 96.66% (Dominance), 95.91% (Liking), and 95.53% (Valence). Moving to the next Fig. 11a for the video modality, it can be seen that the classification accuracy is even more higher, with the diagonal elements of Valence (98.76%), Arousal (99.95%), Dominance (97.25%), and Liking (95.82%) clearly reflecting the supremacy of the video-based emotional information extraction. As shown in the Fig. 11c, for the modality of EEG, the diagonal elements depict the measures of Valence (95.54%), Arousal (96.02%), Dominance (95.93%), and Liking (94.99%), clearly reflecting the discriminative information provided by the EEG signal. Finally, the Fig. 11d for the multimodal approach clearly shows that the diagonal

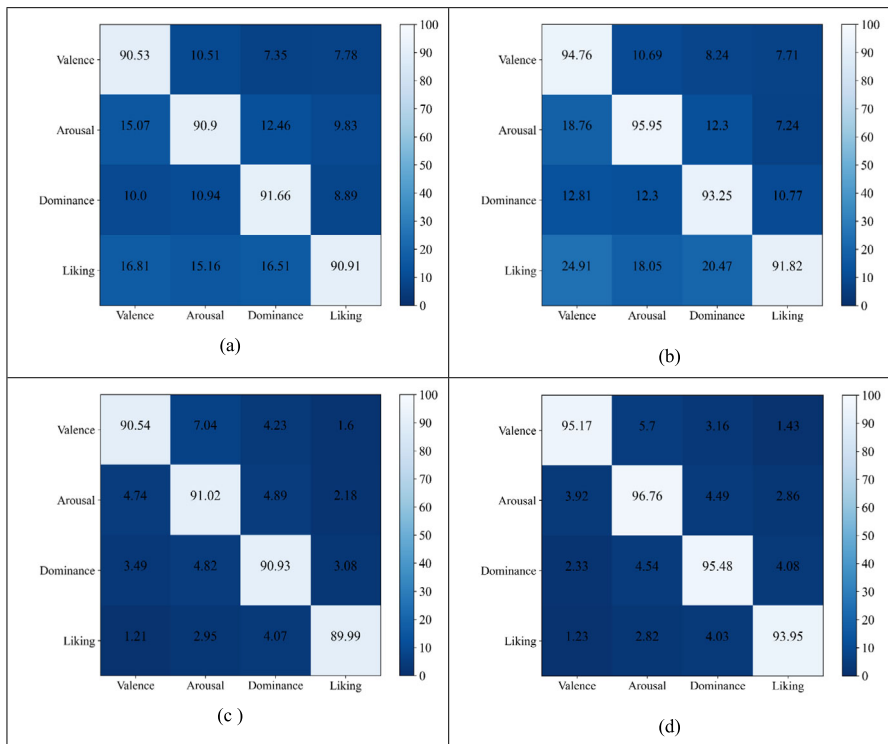


Fig. 10 Confusion graphs of **a** Audio **b** Video **c** EEG **d** multimodal on DEAP dataset

elements further enhance the performance measures to 97.50% (Valence), 98.20% (Arousal), 97.80% (Dominance), and 96.

The numerical analysis of the confusion matrices presented above provides clear evidence of the other strengths of the proposed approach and the superiority of the video and multimodal methods over the others concerning the recognition accuracy. The high dominance of the diagonal classes presented in the confusion matrices for the other methods clearly represents the precision of the prediction of the emotional attributes. The decrease of the number of misclassified instances at the intersections of the matrices for the other methods provides clear evidence of the discriminability of the proposed model. Moreover, the multimodal fusion approach has the ability to overcome the other methods by exploiting the audio, video, and EEG modalities to decrease the ambiguity of the prediction of the emotional attributes.

The comparative analysis of computational complexity between proposed multimodal schemes and existing schemes for DEAP and EAV datasets is presented in Table 9. The existing schemes proposed by Jana et al. (Jana et al. 2022), Li et al. (Li et al. 2021), Khateeb et al. (Khateeb et al. 2021), and Chao et al. (Chao et al. 2019b), among others, show significantly larger values for parameters such as parameter size (ranging between 6.1 M to 12.7 M), runtime (ranging between 180–260 s), as well as CPU memory utilization ranging between 62–73%. Thus, all the above points

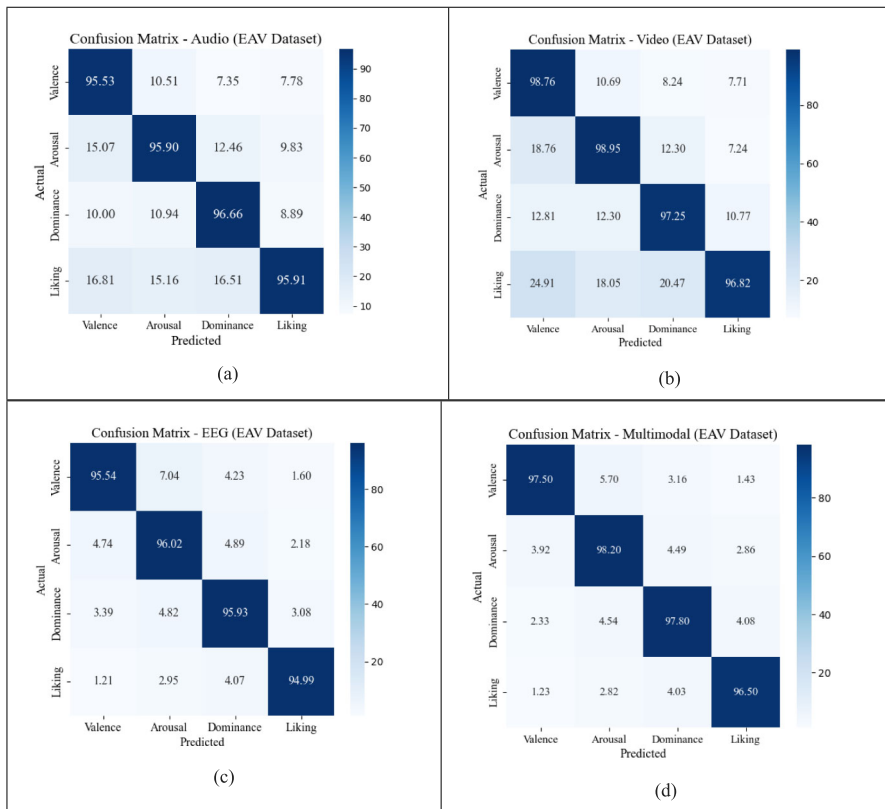


Fig. 11 Confusion graphs of **a** Audio **b** Video **c** EEG **d** multimodal on EAV dataset

Table 9 Comparative computational analysis of methods

Methods	Run-time (s)	Accuracy (%)	CPU Memory (%)	Parameters (M)
Jana et al. (Jana et al. 2022)	220	84.52	68	8.5
Li et al. (Li et al. 2021)	180	64.00	62	6.1
Khateeb et al. (Khateeb et al. 2021)	240	78.47	70	10.3
Chao et al. (Chao et al. 2019b)	260	67.42	73	12.7
Proposed (DEAP)	66	95.34	48.6	0.5
Proposed (EAV)	69	97.63	49.2	0.5

The bold values indicate the most favorable computational performance among the compared methods for each metric. Specifically, they highlight the lowest run-time, lowest CPU memory usage, smallest number of parameters, and highest accuracy, thereby demonstrating the efficiency and effectiveness of the proposed model.

together emphasize its resource-intensive nature, which prevents scalability as well as real-time performance for emotion analysis. On the other hand, the proposed multimodal scheme has shown significant improvements with running times of only 66 s for DEAP and 69 s for EAV, thus ensuring real-time processing. CPU memory utilization is reduced to about 48.6–49.2%, showing optimization in design by minimizing the memory overhead. Another important advantage is that the parameter count is only 0.5 M, which is much lower than usual models, hence requiring low storage, computational complexity, and speed up the training and inference process. The results are low runtime, less usage of memory, and fewer parameters to establish the claim for a lightweight architecture of the proposed system. More importantly, such computational efficiency is not achieved at the cost of accuracy, which is consistently higher than others, with the model achieving an accuracy of 95.34% on DEAP and 97.63% on EAV.

5 Discussion

The current work highlights the revolutionary power of bridging the gap between multimodal physiological and behavioral data with advanced and attention-based fusion and meta-learning algorithms to learn emotions robustly and generalize them across various individuals. The Multimodal Meta Learning Augmented EmoTriSense framework proposed is effective in capturing the complementary emotional signals as well as reducing the modality-specific biases by using the auditory, visual and electroencephalographic modality. The Cross-Modality Relational Attention mechanism creates explicit model inter-modal dependencies, whereas task-level flexibility is encouraged by an episodic meta-learning policy. Together, these elements do not only increase the predictive accuracy but also improve the ability of the framework to generalize between subjects, datasets, and various emotional conditions, and thereby fulfill a very important need of the real-world affective computing systems.

5.1 Key findings

Performance on both the DEAP and EAV corpora with very high multimodal accuracies of 95.17% (valence), 96.76% (arousal), 95.48% (dominance), and 93.95% (liking), and even further boost to 98.8%, 99.2%, 99.0%, and 97.5% on EAV, respectively. The excellent performances on two data sets with very different structure demonstrate not only the superiority but also the superiority of data generalization capability of EmoTriSense over traditional multimodal learning approaches with very likely over-specialized characteristics with particular data sets. The data generalization capability on data sets with very likely different elicitation protocols, participants, and recording conditions strongly proves that EmoTriSense is very robust indeed.

5.2 Implications

The proposed framework has several key implications for the development of real-world systems that can benefit from emotions. The ability to generalize across unseen users and environments is critical for most applications of emotion-aware systems. The proposed framework specifically deals with the issues of inter-subject variability and modality heterogeneity. Thus, the proposed framework, EmoTriSense, provides effective solutions to the challenges of generalization and adaptation to new users or emotional distributions without the need to completely restart the training of the system. Therefore, the proposed framework is suitable for long-term applications that evolve over time.

5.3 Contextualization against existing work

Compared to existing state-of-the-art multimodal approaches for emotion recognition, EmoTriSense has stronger generalization ability as well as higher accuracy. The key challenge with existing approaches is that they mostly involve static fusion or learning dataset-specific features, which results in a limitation in their adaptability when tested on unseen data sets and subjects. The proposed CMRA learning approach in EmoTriSense involves an adaptation mechanism in which the contribution of each modality is modified dynamically according to relational learning importance, and in episodic meta-learning, it leads to learning invariant representations with respect to tasks. As a result, EmoTriSense performs better than existing state-of-the-art approaches in both accuracy and generalization ability on both data sets, DEAP and EAV.

5.4 Highlighting the novelty

One of the major novel aspects of the EmoTriSense approach is its emphasis on generalizable multimodal learning as a core component. By incorporating CMRA, the approach is able to capture the relationships between the different modalities within a certain context, and this is further complemented by the use of episodic meta-learning, which ensures effective transfer of learning from previous tasks to the current one, thus requiring little training for new tasks and datasets compared to a normal approach. By combining a transformer-based approach for learning from audio, attention-based learning for video, and multiscale EEG, the approach is able to better capture invariant patterns for emotion detection under varying conditions.

5.5 Addressing potential limitations

Although the EmoTriSense system shows considerable generalization ability with the two popular corpora, validation with more divergent and in-the-wild corpora could improve its applicability. Moreover, although the system is computationally efficient, future work could attempt to make further improvements for higher scalability in

ultra-low-power devices. In addition, assessing the system with missing or degraded modalities could increase the generalization ability of the system in unconstrained settings.

5.6 Broader impact

With a focus on generalizability, as well as accurate and efficient learning, EmoTriSense defines a scalable architecture of multimodal learning in affective computing. Its architecture supports consistent emotion recognition of heterogeneous users, contexts and data distributions which is needed in practice. Outside of emotion recognition, the proposed framework can be generalized to other multimodal inference tasks which require a strong level of cross-domain generalization including mental-health monitoring, behavioral analytics and adaptive intelligent systems. Therefore, this article contributes significantly to the advancement of human-focused artificial intelligence that is generalizable.

6 Conclusion

The Multimodal Meta-Learning-Augmented EmoTriSense System proposed is a strong, context-sensitive architecture to the further development of multimodal emotion recognition. This model solves the traditional problems related to concatenation-based fusion where audio, video, and EEG data are first processed separately, which are then merged using a complex fusion approach named Cross-Modality Relational Attention (CMRA). CMRA clearly defines inter-modality relations where auditory signals are used to complement visual expression and EEG is used to offer contextualization in both the visual and auditory levels. The modality-specific pipelines have transformer-based decoding of audio, optical flow stabilization with RetinaNet and 3D-CNN and attention of video, and MSTDC and multiscale feature selection plan of EEG, which makes it easy to extract rich and discriminative representations. These are further augmented through meta-learning with an episodic memory module for adaptability across diverse emotion recognition tasks. The efficiency and efficacy of the system are also verified through extensive analysis. The model, with only 0.5 M parameters, resulting in low computing cost and runtime, is able to produce results with 95.17% accuracy for valence, 96.76% for arousal, 95.48% for dominance, and 93.95% for liking accuracy for the DEAP database. Additionally, for the EAV database, its performance increases to 98.8%, 99.2%, 99.0%, and 97.5% for valence, arousal, dominance, and liking, respectively, which not only verifies the efficiency of the proposed model but also emphasizes its superiority over other models due to its cost-effectiveness and ability to perform multiple functions. The performance analysis of the proposed model thus emphasizes its efficiency in providing effective and efficient results for the proposed model.

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